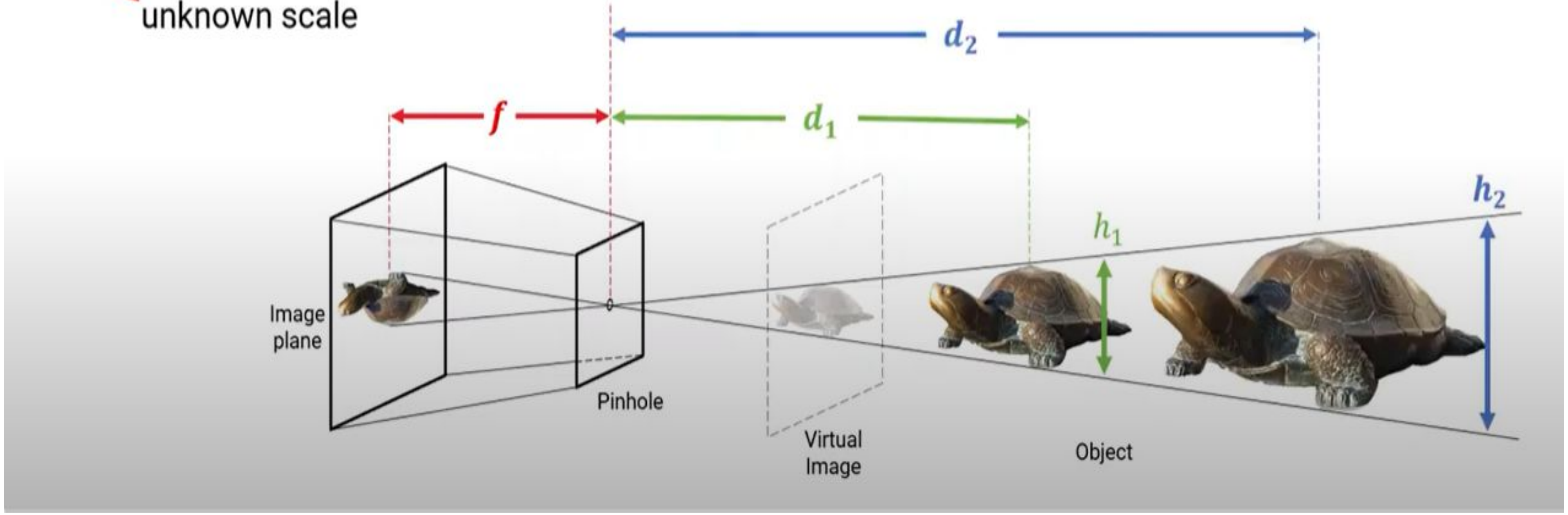
Stereopsis

- Matrix for projection equation
- Projecting 3D world coordinates in to pixel coordinates

$$\lambda \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{pmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$$

$$\lambda \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{pmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$$

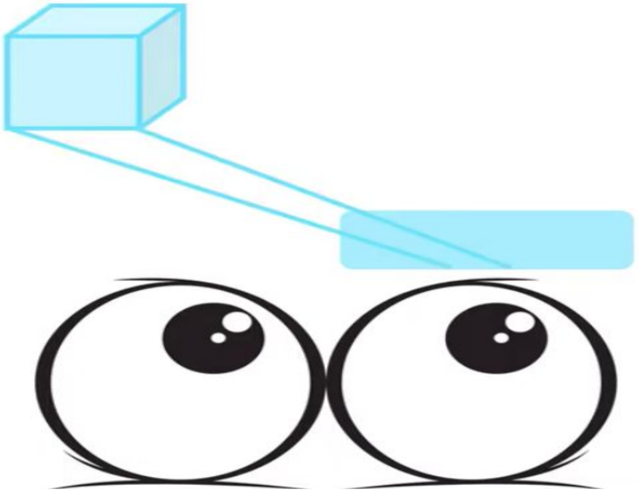
unknown scale

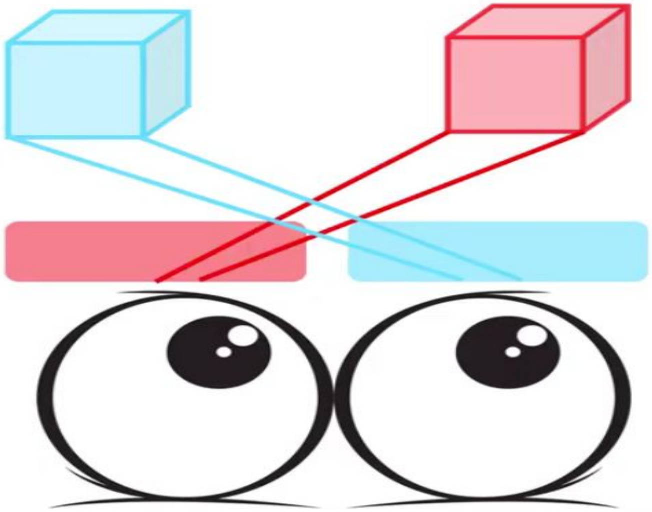


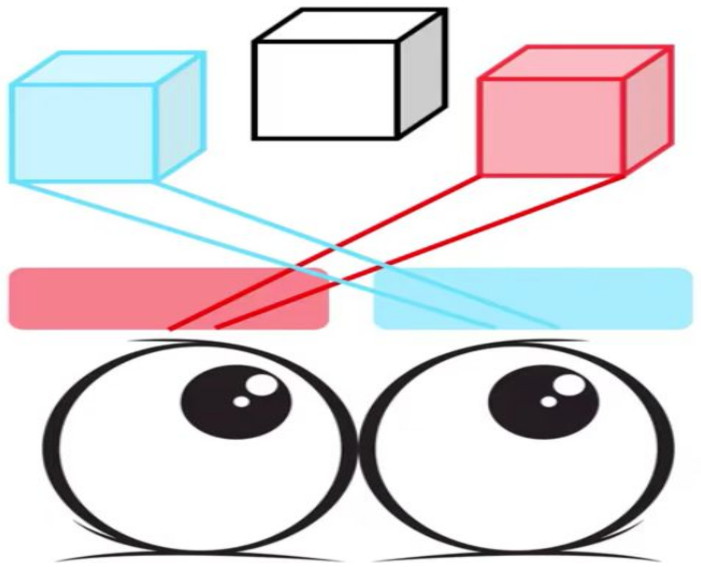
Stereoscopic Image


CRASH

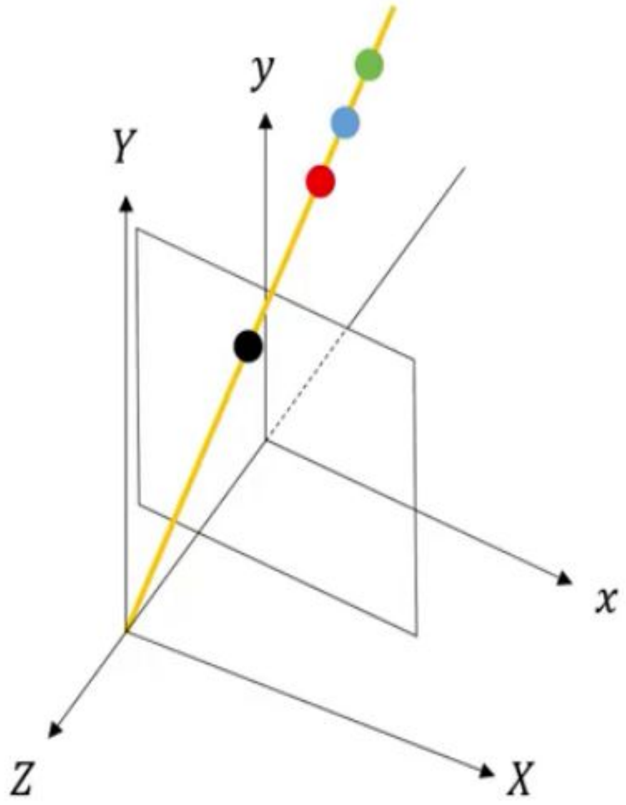
3D Glasses





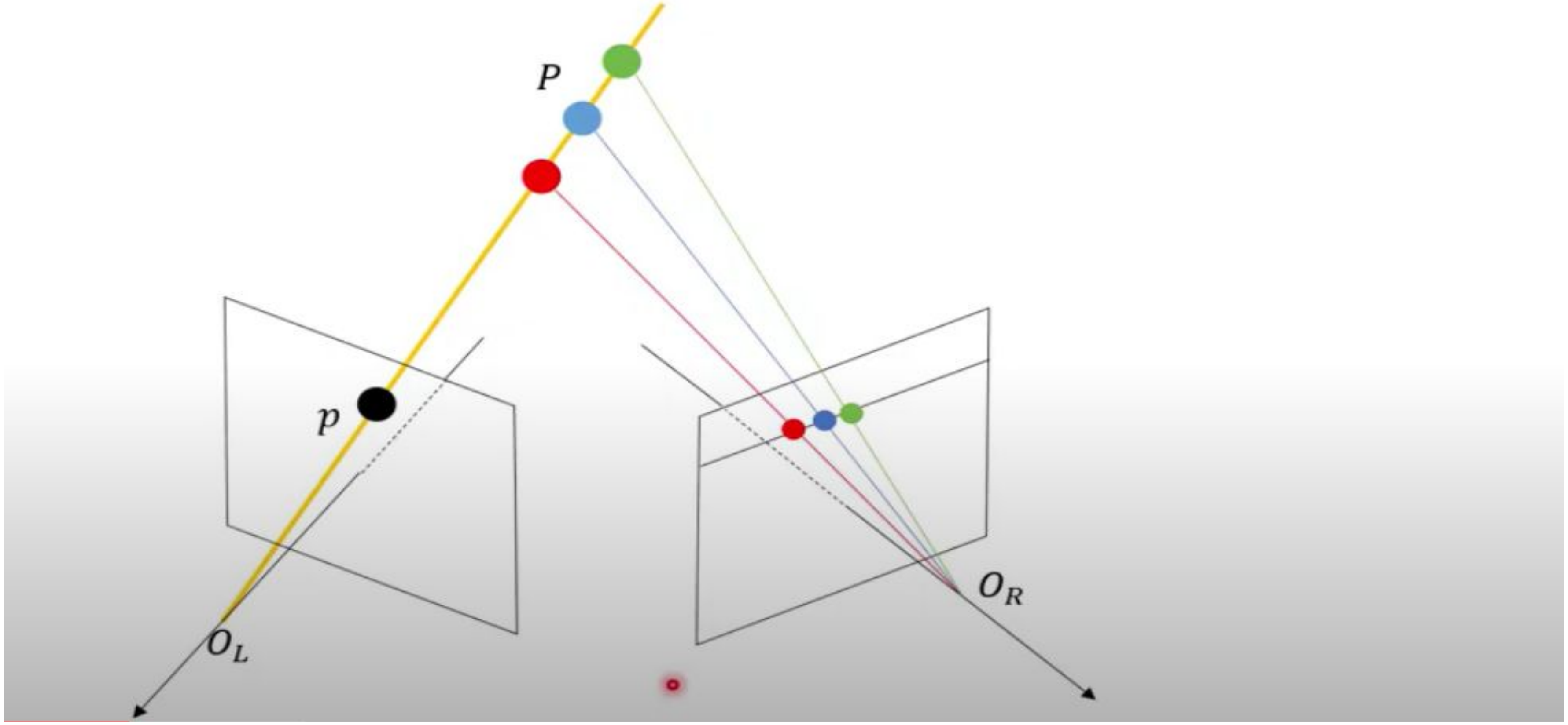


Stereo vision



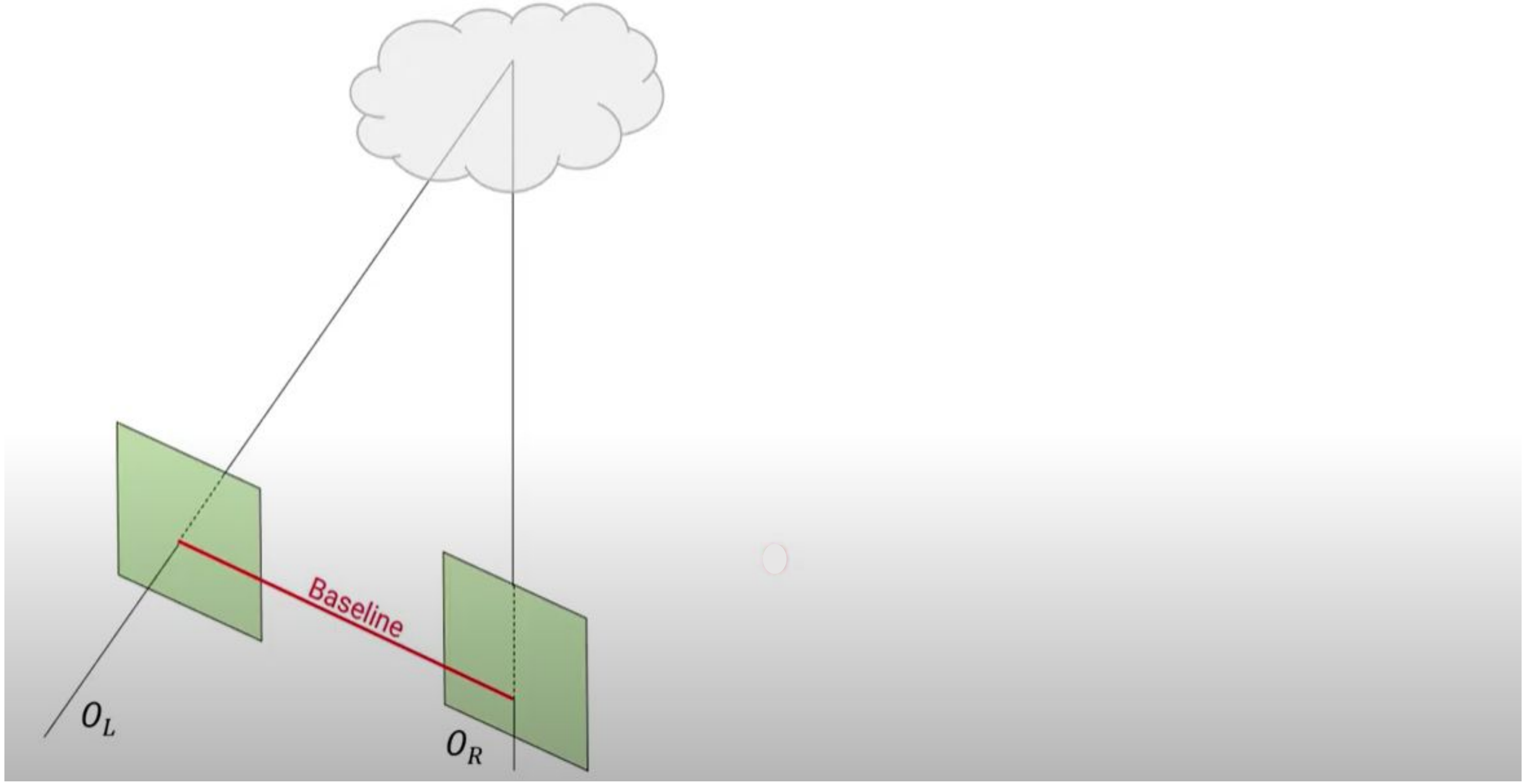
$$x = f\frac{X}{Z} = f\ \frac{kX}{kZ} = f\frac{X_1}{Z_1} = f\frac{X_2}{Z_2} = f\frac{X_3}{Z_3}$$

Adding another camera



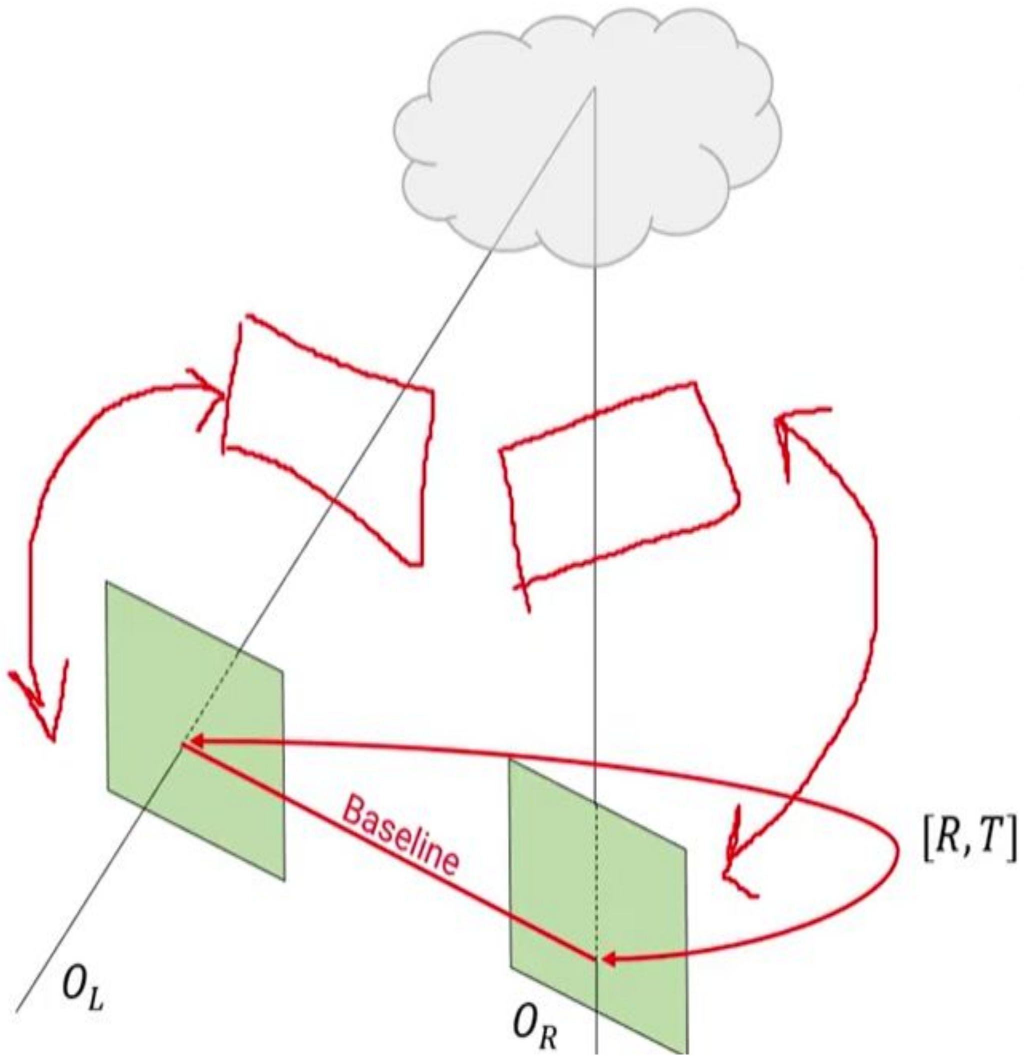
Left Right

- For each pixel in the first image
  - Find corresponding epipolar line in the right image
  - Examine all pixels on the epipolar line and choose the best match
  - Triangulate the matches to estimate depth of the 3D world point



Baseline
O_L
O_R

Simplest Case – parallel camera

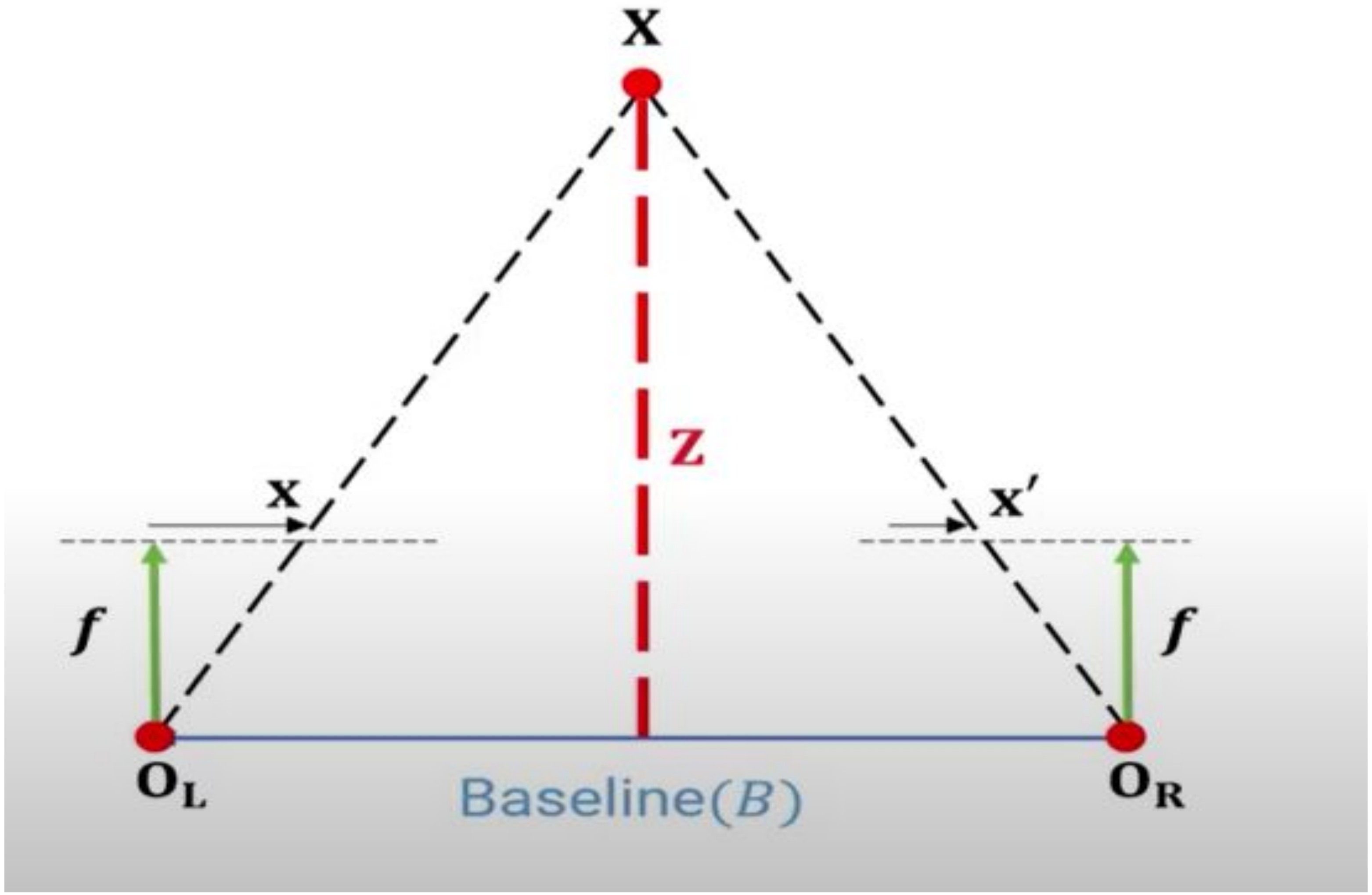


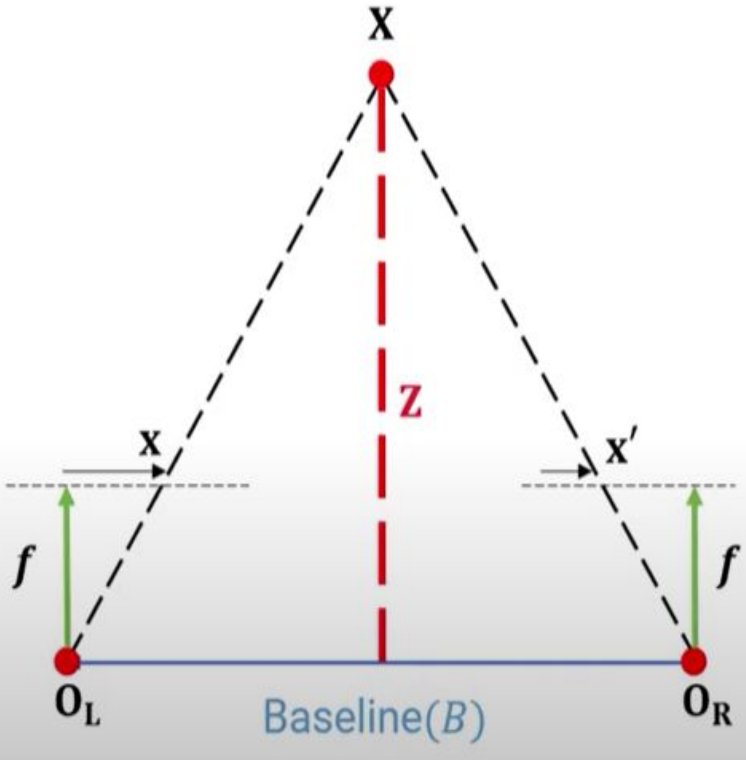
- Assume that image planes of cameras are parallel to each other and to the baseline
- Focal lengths are same

⇒ Epipolar lines fall along the horizontal lines of the two images

- Known rotation and translation!
- Calibrated Stereo pair (or known extrinsics)

Depth from disparity





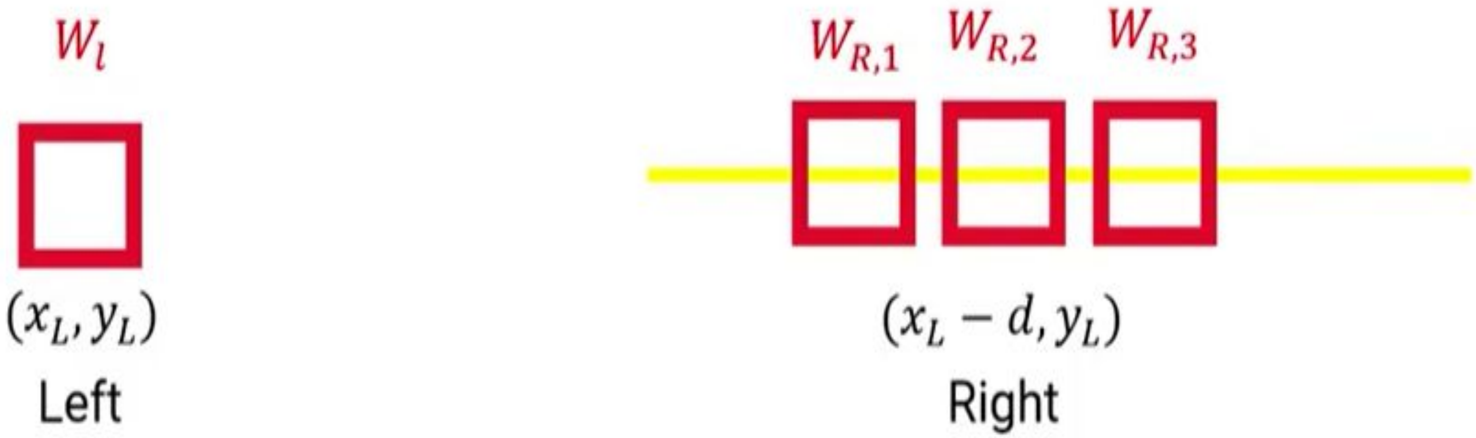
$$\frac{x - x'}{O_L - O_R} = \frac{f}{Z}$$

$$disparity(d) = x - x' = B \cdot \frac{f}{Z}$$

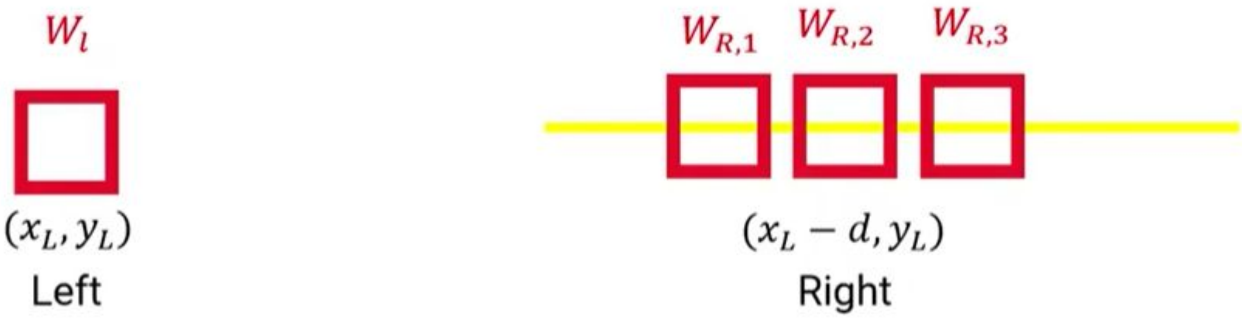
Stereo matching algorithm

- For each pixel in the first image
  - Find corresponding epipolar line in the right image
  - Examine all pixels on the epipolar line and choose the best match
  - Triangulate the matches to estimate depth of the 3D world point

Sum of Squared Difference



Which one is the best match for W_l?



Which one is the best match for W_l?

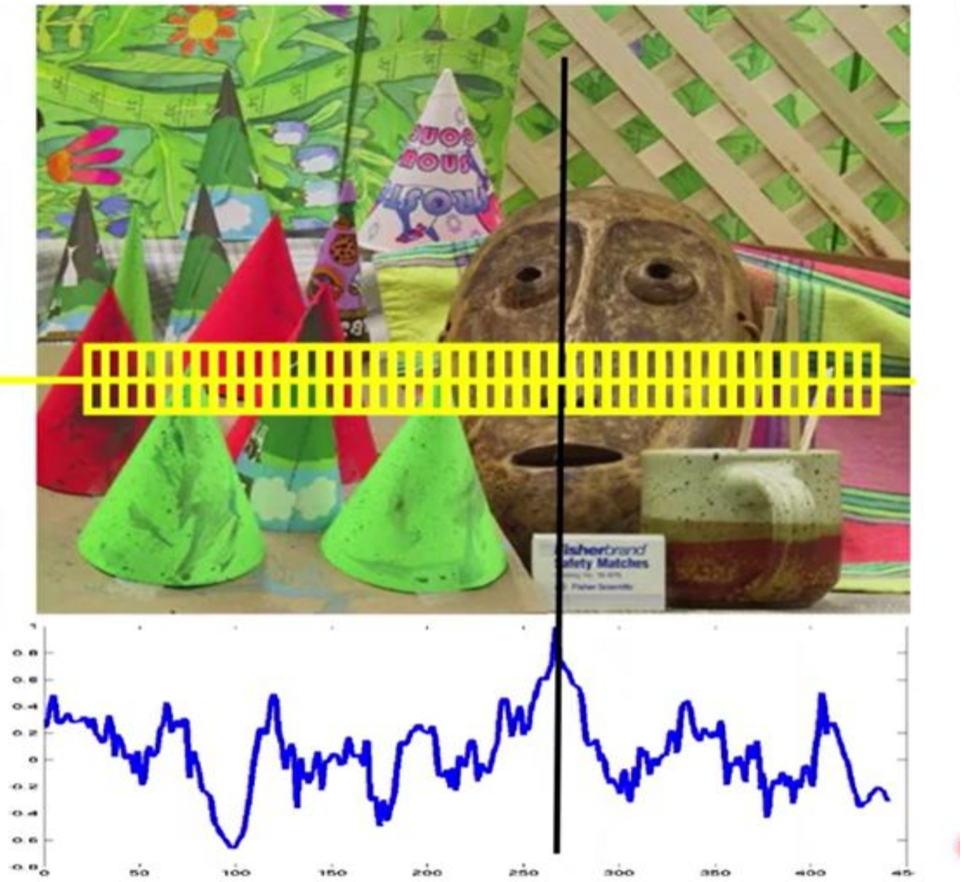
Compute SSD cost for each window

- It measures the intensity difference as a function of disparity

$$C_r(x, y, d) = \Sigma_{\forall(u,v)\in W_m(x,y)} \; [I_L(u, v) - I_R(u - d, v)]^2$$

cost

Window sizes are user-defined!



SSD score vs $x - axis$ along the epipolar line

- Repeat this for all pixels /window
- So the window size should not be too small or too big

Binocular Fusion

- Binocular vision is vision in which both eyes are used together.
- We have two eyes, each of which forms its own image.
- Yet we do not see an object double because our brain fuses the two images into one.
- Thus we possess 'binocular vision'.
- It is because of good fusion that our vision is three dimensional and we are able to locate an object in space accurately.